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Electronic device including flexible display and method for controlling input area for expansion and reduction of display in the electronic device

Abstract

An electronic device includes: a first housing, a second housing for accommodating the first housing to enable sliding of the first housing, a flexible display connected to the first housing, a slide driving unit for slidably moving the first housing, and a processor. The processor identifies a reduced state where the flexible display is slid in a reduction direction or an expanded state where the flexible display is slid in an expansion direction, in the reduced state, identifies a first edge area of the flexible display as a first input area and controls the flexible display to enter the expanded state, based on an input through the first input area, and in the expanded state, identifies a second edge area of the flexible display as a second input area and controls the flexible display to enter the reduced state, based on an input through the second input area.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11043191	12/2020	Kang	N/A	G09G 5/10
11184987	12/2020	Jung	N/A	H05K 5/0217
12034879	12/2023	Kushwaha	N/A	H04M 1/0268
12035035	12/2023	Kumar Agrawal	N/A	H04N 23/631
2010/0167791	12/2009	Lim	455/566	G06F 1/1624
2016/0373654	12/2015	Kwon	N/A	G09G 5/373
2017/0011714	12/2016	Eim	N/A	G06F 1/1677
2018/0348881	12/2017	Chung	N/A	H04W 88/02
2019/0261519	12/2018	Park	N/A	G06F 1/1652
2020/0076940	12/2019	Kim	N/A	G06F 1/1656
2020/0105169	12/2019	Jeong	N/A	H10K 77/111
2020/0264673	12/2019	Kim	N/A	G06F 1/1652
2020/0264674	12/2019	Km	N/A	G06F 1/1681
2021/0201721	12/2020	Li	N/A	G09G 3/035
2021/0201722	12/2020	Jeong	N/A	G09F 9/37
2021/0247805	12/2020	Min	N/A	H04M 1/0243
2021/0278878	12/2020	Shim	N/A	G06F 1/1656
2021/0352813	12/2020	Cho	N/A	G06F 1/1656
2022/0113842	12/2021	Kim	N/A	H04M 1/0268
2022/0148464	12/2021	Kwak	N/A	G06F 1/1643
2022/0148473	12/2021	Kim	N/A	G06F 1/1652

2022/0335869	12/2021	Kwak	N/A	G06F 1/1624
2022/0368784	12/2021	Koh	N/A	G06F 1/16
2022/0383786	12/2021	Kwak	N/A	G06F 1/1643
2022/0390983	12/2021	Gao	N/A	G09F 9/30
2023/0007111	12/2022	Lee	N/A	H04M 1/0268
2023/0021496	12/2022	Choi	N/A	H01Q 1/521
2023/0039953	12/2022	Seger, Jr	N/A	G06F 3/0416
2023/0093986	12/2022	Eom	361/679.27	H04M 1/0235
2023/0109078	12/2022	Li	345/173	G06F 1/1626
2023/0110166	12/2022	Chung	361/679.01	G06F 1/1677
2023/0122508	12/2022	Kim	318/558	G06F 1/1679
2023/0130795	12/2022	Yoon	361/807	H04M 1/0268
2023/0176720	12/2022	Cheon	345/173	G06F 1/1624
2023/0389243	12/2022	Cho	N/A	H05K 7/20963
2024/0126325	12/2023	Kumar Agrawal	N/A	G06F 1/1677
2024/0129394	12/2023	Vashist	N/A	H04M 1/0241
2024/0192726	12/2023	Manning	N/A	G06F 1/1618
2024/0196550	12/2023	Cho	N/A	G06F 1/1624
2024/0203310	12/2023	Noh	N/A	G06F 1/1681
2024/0223686	12/2023	Kim	N/A	H04M 1/0241
2024/0302883	12/2023	Prabhu	N/A	G06F 1/1677
2024/0310878	12/2023	Pailla	N/A	G06F 1/1652
2024/0385660	12/2023	Zeng	N/A	H04M 1/0268
2025/0008672	12/2024	Lee	N/A	H01Q 1/243
2025/0030789	12/2024	Cho	N/A	H04M 1/0237

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-2016-0013289	12/2015	KR	N/A
10-2016-0092776	12/2015	KR	N/A
10-1784880	12/2016	KR	N/A
10-2256042	12/2020	KR	N/A
10-2021-0113805	12/2020	KR	N/A
20210146095	12/2020	KR	N/A
20220014751	12/2021	KR	N/A
20220077062	12/2021	KR	N/A
20230014279	12/2022	KR	N/A
2021-045275	12/2020	WO	N/A
2021-095925	12/2020	WO	N/A
2021-160276	12/2020	WO	N/A

OTHER PUBLICATIONS

International Search Report mailed Jan. 12, 2023 for PCT/KR2022/015285. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of International Application No. PCT/KR2022/015285 designating the United States, filed on Oct. 11, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2021-0135185, filed on Oct. 12, 2021, in the Korean Intellectual Property Office, and to Korean Patent Application No. 10-2022-0013163, filed on Jan. 28, 2022, in the Korean Intellectual Property Office, the disclosures of all of which are incorporated by reference herein in their entireties

TECHNICAL FIELD

(1) Various embodiments relate to an electronic device including a flexible display.

BACKGROUND ART

(2) In line with development of electronics, information, and communication technologies, various functions are integrated into a single portable communication device or electronic device. For example, a smartphone includes not only a communication function, but also the function of a music player, an imaging device, or an electronic wallet, and more diversified functions may be implemented in the smartphone by installing additional applications.

(3) In line with widespread use of personal or portable communication devices such as smartphones, there have been increasing user demands for portability and use convenience. For example, a touchscreen display is an output device for outputting visual information (for example, screen), and may also provide a virtual keypad which replaces a mechanical input device (for example, button-type input device). As a result, the portable communication device or electronic device has become able to provide identical or more improved utility (for example, larger screen) while having a compact size. On the other hand, it is expected that commercialization of flexible displays, for example, foldable or rollable display, will further improve the portability and use convenience of electronic devices.

DISCLOSURE

Technical Problem

(4) An electronic device including a flexible display may be implemented such that structures of the electronic device can relatively move (for example, slide, spin, or rotate) with respect to each other. For example, the flexible display may provide an extension structure following a sliding movement when some structures (for example, a first housing and a partial area of the flexible display) has moved into or away from another structure (for example, a second housing). For example, the flexible display may perform a slide-in or slide-out operation (for example, sliding movement) in the electronic device, thereby expanding or reducing the display (or display area). The electronic device may automatically perform a side-in or slide-out operation of the flexible display based on a designated trigger (for example, user input).

(5) The electronic devices may automatically perform side-in or slide-out operations of flexible displays in various user input types, including a physical button input type, a software button input type, and a gesture input type. However, conventional user input types simply use physical or software-based button touches (or pressing) like other input types, or use gestures which are also used by other functions, and user may thus need to recognize and learn the input type, or may be confused even after learning the same.

Technical Solution

(6) Various embodiments of the disclosure may provide an electronic device including a flexible display, wherein an input for sliding of the flexible display (or expansion and reduction of the display) can be performed more intuitively and conveniently, and a method for controlling an input area for display expansion and reduction in the electronic device including a flexible display.

(7) Various embodiments of the disclosure may provide an electronic device including a flexible display, wherein an input area is configured in the sliding direction of the flexible display (for example, the direction in which the display is expanded or reduced), and inputs can be performed

through the configured input area such that the user can easily make an input for expanding or reducing the display in an intuitive manner, an input for sliding of the flexible display (or expansion and reduction of the display) can be performed more intuitively and conveniently, and a method for controlling an input area for display expansion and reduction in the electronic device including a flexible display.

(8) According to an embodiment of the disclosure, an electronic device includes a first housing, a second housing configured to accommodate at least a portion of the first housing and connected to the first housing to enable sliding of the first housing, a flexible display connected to the first housing, a slide driving unit configured to slidably move the first housing and the flexible display in an expansion direction or a reduction direction opposite to the expansion direction, and at least one processor. The at least one processor is configured to, identify a reduced state in which the flexible display is slid in the reduction direction or an expanded state in which the flexible display is slid in the expansion direction, in the reduced state, identify a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and control the slide driving unit such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identify a second edge area of the flexible display disposed in the expansion direction from the center of the flexible display as a second input area and control the slide driving unit such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

(9) According to an embodiment of the disclosure, a method for controlling an input area for expansion and reduction of a display in an electronic device may include identifying a reduced state in which a flexible display is slid in a reduction direction or an expanded state in which the flexible display is slid in an expansion direction, in the reduced state, identifying a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and performing control such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identifying a second edge area of the flexible display disposed in the expansion direction from the center of the flexible display as a second input area and performing control such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

(10) According to an embodiment of the disclosure, in connection with a non-volatile storage medium storing instructions, the instructions may, when executed by at least one process, cause an electronic device comprising at least one processor to perform at least one operation, the at least one operation including identifying a reduced state in which a flexible display is slid in a reduction direction or an expanded state in which the flexible display is slid in an expansion direction, in the reduced state, identifying a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and performing control such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identifying a second edge area of the flexible display disposed in the expansion direction from the center of the flexible display as a second input area and performing control such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

(11) According to an embodiment of the disclosure, an electronic device includes: a first housing; a second housing configured to accommodate at least a portion of the first housing and including an open side through which the first housing is slidable; a flexible display connected to the first housing; a slide driving unit configured to slidably move the first housing and the flexible display in an expansion direction or a reduction direction opposite the expansion direction; and at least one processor. The at least one processor is configured to identify a reduced state in which the flexible display is slid in the reduction direction or an expanded state in which the flexible display is slid in the expansion direction, in the reduced state, identify a first edge area of the flexible display corresponding to a sidewall of the second housing as a first input area and control the slide driving

unit such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identify a second edge area of the flexible display corresponding to an outermost sidewall of the first housing as a second input area and control the slide driving unit such that the flexible display enters the reduced state, based on reception of a second input through the second input area. The sidewall of the second housing is disposed opposite to the open side, and the outermost sidewall of the first housing is disposed opposite to the sidewall of the second housing.

Advantageous Effects

(12) According to various embodiments of the disclosure, an input for sliding of a flexible display (or expansion and reduction of the display) can be performed more intuitively and conveniently in an electronic device.

(13) According to various embodiments of the disclosure, an input area is configured in the sliding direction of a flexible display (for example, the direction in which the display is expanded or reduced), and inputs can be performed through the configured input area such that the user can make an input for expanding or reducing the display in an intuitive and easy manner in an electronic device.

(14) Advantageous effects obtainable from the disclosure may not be limited to the above mentioned effects, and other effects which are not mentioned may be clearly understood, through the following descriptions, by those skilled in the art to which the disclosure pertains.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an electronic device in a network environment according to one embodiment;

(2) FIG. 2A illustrates a state in which a second display area of a flexible display is accommodated in a second housing according to one embodiment;

(3) FIG. 2B illustrates a state in which a second display area of a flexible display is exposed to the outside of a second housing according to one embodiment;

(4) FIG. 3 illustrates a sensing structure for sensing expansion or contraction of a flexible display according to one embodiment;

(5) FIG. 4 is a block diagram of an electronic device according to one embodiment;

(6) FIG. 5 is a flowchart illustrating an input area control operation for expanding and reducing a display in an electronic device including a flexible display according to one embodiment;

(7) FIG. 6A illustrates an example in which an electronic device transitions to an expanded state from a reduced state according to one embodiment;

(8) FIG. 6B illustrates an example in which an electronic device transitions to a reduced state from an expanded state according to one embodiment;

(9) FIG. 7A illustrates GUI guide information displayed in a first input area in a reduced state of an electronic device according to one embodiment;

(10) FIG. 7B illustrates GUI guide information displayed in a second input area in an expanded state of an electronic device according to one embodiment;

(11) FIG. 8A illustrates an example in which an electronic device transitions to an expanded state from a reduced state when the electronic device is in a landscape view mode, according to one embodiment;

(12) FIG. 8B illustrates an example in which an electronic device transitions to a reduced state from an expanded state when the electronic device is in a landscape view mode, according to one embodiment;

(13) FIG. 9 illustrates an example in which a first input or a second input designated to be

identified in a first input area or a second input area of an electronic device is a swipe input after a long touch, according to one embodiment;

(14) FIG. 10 is a flowchart showing an input area control operation for full expansion, partial expansion, and reduction of a display in an electronic device according to one embodiment;

(15) FIG. 11 illustrates a reduced state, a partially expanded state, and a fully expanded state of a display in an electronic device according to one embodiment;

(16) FIG. 12 illustrates an example in which a first edge area in a first direction of a display is designated as a first input area, and a second edge area in a second direction is designated as a second input area of a display according to one embodiment;

(17) FIG. 13 illustrates an example in which an electronic device enters a partially expanded state or a fully expanded state during display of a content while the electronic device is in a reduced state and in a landscape view mode, according to one embodiment;

(18) FIG. 14 illustrates an example of utilizing an expanded area when an electronic device transitions to a partially expanded state or a fully expanded state from a reduced state according to one embodiment;

(19) FIG. 15 is a flowchart showing a control operation when a third edge area at the lower end of a display is used as an input area for expanding or reducing the display in an electronic device according to one embodiment; and

(20) FIG. 16 illustrates an example of receiving a first input or a second input through a third edge area at the lower end of a display according to one embodiment.

(21) In connection with the description of the drawings, the same or similar reference numerals may be used for the same or similar elements.

MODE FOR INVENTION

(22) The terms used in the disclosure are only used to describe specific embodiments, and are not intended to limit the disclosure. A singular expression may include a plural expression unless they are definitely different in a context. Unless defined otherwise, all terms used herein, including technical and scientific terms, have the same meaning as those commonly understood by a person skilled in the art to which the disclosure pertains. Such terms as those defined in a generally used dictionary may be interpreted to have the meanings equal to the contextual meanings in the relevant field of art, and are not to be interpreted to have ideal or excessively formal meanings unless clearly defined in the disclosure. In some cases, even the term defined in the disclosure should not be interpreted to exclude embodiments of the disclosure.

(23) FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to an embodiment.

(24) Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(25) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101**

coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(26) The auxiliary processor **123** may control, for example, at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active (e.g., executing an application) state. According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence model is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(27) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(28) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(29) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(30) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(31) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(32) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., an electronic device **102** (e.g., a speaker or a headphone)) directly or wirelessly coupled with the electronic device **101**.

(33) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(34) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(35) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

(36) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(37) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(38) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(39) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(40) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation

satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(41) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing 1 eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(42) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(43) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(44) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile

industry processor interface (MIPI)).

(45) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(46) FIG. 2A illustrates a state in which a second display area of a flexible display is accommodated in a second housing according to one embodiment. FIG. 2B illustrates a state in which a second display area of a flexible display is exposed to the outside of a second housing according to one embodiment.

(47) FIGS. 2A and 2B illustrate a structure expanding in the right direction of the flexible display **203** when viewed from the front of the electronic device **101**. The state shown in FIG. 2A may be defined as a state in which the first housing **201** is closed with respect to the second housing **202** (hereinafter “closed state”), and the state shown in FIG. 2B may be defined as a state in which the first housing **201** is opened with respect to the second housing **202** (hereinafter “open state”). According to an embodiment, a “closed state” or an “open state” may be defined as a state in which the electronic device is closed or opened.

(48) Referring to FIGS. 2A and 2B, the electronic device **101** may include housings **201** and **202**. The housings **201** and **202** may include a second housing **202** and a first housing **201** movably disposed with respect to the second housing **202**. In an embodiment, the electronic device **101** may be interpreted as having a structure in which the second housing **202** is slidably disposed on the first housing **201**. According to one embodiment, the first housing **201** may be arranged to reciprocate by a predetermined distance in a direction shown with reference to the second housing **202**, for example, directions indicated by arrow {circle around (1)}. The configuration of the electronic device **101** of FIGS. 2A and 2B may be entirely or partially the same as that of the electronic device **101** of FIG. 1.

(49) According to one embodiment, the first housing **201** may be referred to as, for example, a first structure, a slide unit, or a slide housing, and may be reciprocally disposed on the second housing **202**. According to one embodiment, the first housing **201** may accommodate various electrical and electronic components such as a circuit board or a battery. The second housing **202** may be referred to as, for example, a second structure, a main unit, or a main housing, and may guide the movement of the first housing **201**. A portion of the flexible display **203** (e.g., the first display area **A1**) may be seated in the first housing **201**. According to one embodiment, according to the movement (e.g.,

sliding movement) of the first housing **201** with respect to the second housing **202**, another portion of the flexible display **203** (e.g., the second display area **A2**) may be accommodated (e.g., slide-in operation) inside the second housing **202** or exposed to the outside (e.g., slide-out operation) of the second housing **202**. According to one embodiment, a motor, a speaker, a SIM socket, and/or a sub circuit board electrically connected to the main circuit board may be disposed in the first housing **201**. A main circuit board on which electrical components, such as an application processor (AP) and a communication processor (CP), are mounted may be disposed in the second housing **202**.

(50) According to one embodiment, the first housing **201** may include a first plate **211** (e.g., a slide plate). The first plate **211** may include a first face (e.g., the first face **F1** of FIG. 2B) forming at least a portion of the first plate **211**, and a second face **F2** facing in the opposite direction of the first face **F1**. According to one embodiment, the first plate **211** may support at least a portion of the flexible display **203** (e.g., the first display area **A1**). According to one embodiment, the first housing **201** may include the first plate **211**, a 1-1.sup.th sidewall **211a** (in other word, outermost sidewall) extending from the first plate **211** in an expansion direction, a 1-2.sup.th sidewall **211b** extending from the 1-1.sup.th sidewall **211a** and the first plate **211** in a top direction, and a 1-3.sup.th sidewall **211c** extending from the 1-1.sup.th sidewall **211a** and the first plate **211** in a bottom direction and parallel to the 1-2.sup.th sidewall **211b**.

(51) According to one embodiment, the second housing **202** may include a second plate (a main case) **221**, a 2-1.sup.th sidewall **221a** extending from the second plate **221** in a reduction direction, a 2-2.sup.th sidewall **221b** extending from the 2-1.sup.th sidewall **221a** and the second plate **221** in a bottom direction, and a 2-3.sup.th sidewall **221c** extending from the 2-1.sup.th sidewall **221a** and the second plate **221** in a top direction and parallel to the 2-2.sup.th sidewall **221b**. According to one embodiment, the 2-2.sup.th sidewall **221b** and the 2-3.sup.th sidewall **221c** may be formed to be perpendicular to the 2-1.sup.th sidewall **221a**. According to one embodiment, the second plate **221**, the 2-1.sup.th sidewall **221a**, the 2-2.sup.th sidewall **221b**, and the 2-3.sup.th sidewall **221c** may form a shape having one open side (e.g., a front face) so as to accommodate (or surround) at least a portion of the first housing **201**. For example, the first housing **201** may be coupled to the second housing **202** to be at least partially surrounded thereby, and may slide in a direction parallel to the first face **F1** and the second face **F2**, for example, in the directions indicated by arrow {circle around (1)} through the open side of the second housing **202**, while being guided by the second housing **202**. According to one embodiment, the second plate **221**, the 2-1.sup.th sidewall **221a**, the 2-2.sup.th sidewall **221b**, and/or the 2-3.sup.th sidewall **221c** may be integrally formed with each other. According to another embodiment, the second plate **221**, the 2-1.sup.th sidewall **221a**, the 2-2.sup.th sidewall **221b**, and/or the 2-3.sup.th sidewall **221c** may be formed as separate housings to be combined or assembled with each other.

(52) According to one embodiment, the second plate **221** and/or the 2-1.sup.th sidewall **221a** may cover at least a portion of the flexible display **203**. For example, at least a portion of the flexible display **203** may be accommodated inside the second housing **202**, and the second plate **221** and/or the 2-1.sup.th sidewall **221a** may cover a portion of the flexible display **203** accommodated inside the second housing **202**.

(53) According to various embodiments, the first housing **201** may be movable in a direction (e.g., directions indicated by the arrow {circle around (1)}) parallel to the 2-2.sup.th sidewall **221b** and the 2-3.sup.th sidewall **221c** to be in an opened state and a closed state with respect to the second housing **202**. The first housing **201** may move to be positioned at a first distance from the 2-1.sup.th sidewall **221a** in the closed state (i.e., reduced state) and to be positioned at a second distance greater than the first distance from the 2-1.sup.th sidewall **221a** in the opened state (i.e., expanded state). In an embodiment, in the closed state, the first housing **201** may surround a portion of the 2-1.sup.th sidewall **221a**.

(54) According to one embodiment, the electronic device **101** may include a flexible display **203**, a key input device **241**, a connector hole **243**, audio modules **247a** and **247b** (not shown), or a

camera module **249a** and **249b** (not shown). Although not shown, the electronic device **101** may further include an indicator (e.g., an LED device) or various sensor modules. The configuration of the flexible display **203**, the audio modules **247a** and **247b** (not shown), and the camera modules **249a** and **249b** (not shown) of FIGS. 2A and 2B may be entirely or partially the same as that of the display module **160**, the audio module **170**, and the camera module **180** of FIG. 1.

(55) According to various embodiments, the flexible display **203** may include a first display area **A1** and a second display area **A2**. According to one embodiment, the first display area **A1** may be disposed on the first housing **201**. For example, the first display area **A1** may extend substantially across at least a portion of the first face **F1** to be disposed on the first face **F1**. The second display area **A2** may extend from the first display area **A1** and may be inserted or accommodated inside the second housing **202** (e.g., a structure) or may be exposed to the outside of the second housing **202** according to the sliding movement of the first housing **201**.

(56) According to various embodiments, the second display area **A2** may move while being substantially guided by one area of the first housing **201**, so as to be accommodated in a space formed inside the second housing **202** or between the first housing **201** and the second housings **202** or may be exposed to the outside. According to one embodiment, the second display area **A2** may move based on the sliding movement of the first housing **201** in the direction (e.g., the directions indicated by arrow {circle around (1)}). For example, while the first housing **201** slides in the reduction direction, a portion of the second display area **A2** may be deformed into a curved shape at a position corresponding to a curved surface of the first housing **201**.

(57) According to various embodiments, in case that the first housing **201** moves from the closed state to the open state when viewed from the top of the first plate **211** (e.g., a slide plate), the second display area **A2** may be gradually exposed to the outside of the second housing **202** and form a substantially flat surface together with the first display area **A1**. The flexible display **203** may be coupled to or disposed adjacent to a touch sensing circuit, a pressure sensor capable of measuring the intensity (pressure) of a touch, and/or a digitizer that detects a magnetic field type stylus pen. In one embodiment, the second display area **A2** may be at least partially accommodated inside the second housing **202**, and even in the state (e.g., the closed state) shown in FIG. 2A, a portion of the second display area **A2** may be visually exposed to the outside. According to one embodiment, irrespective of the closed state or the opened state, a portion of the exposed second display area **A2** may be positioned on a portion of the first housing, and a portion of the second display area **A2** may be maintained in a curved surface.

(58) According to one embodiment, the electronic device **101** may include at least one hinge structure (not shown). The hinge structure may connect the first housing **201** and the second housing **202** to each other. For example, the hinge structure may be connected to the first plate **211** and the second plate **221**. According to one embodiment, the hinge structure may transmit a driving force for guiding the sliding movement of the first housing **201** to the first housing **201**. For example, the hinge structure may include an elastic material (e.g., a spring), and may provide an elastic force in the direction (e.g., directions indicated by the arrow {circle around (1)} of FIG. 2B), based on the sliding movement of the first housing **201**. According to one embodiment, the hinge structure may be excluded.

(59) According to one embodiment, the key input device **241** may be positioned in one area of the first housing **201**. Depending on the appearance and usage state, the electronic device **101** may be designed such that illustrated key input device **241** is omitted therefrom or an additional key input device(s) is included therein. According to one embodiment, the electronic device **101** may include a key input device which is not shown herein, for example, a home key button or a touch pad disposed around the home key button. According to another embodiment, at least some of the key input devices **241** may be disposed on the 2-1.sup.th sidewall **221a**, the 2-2.sup.th sidewall **221b**, or the 2-3.sup.th sidewall **221c** of the second housing **202**.

(60) According to various embodiments, the connector hole **243** may be omitted according to an

embodiment, and may accommodate a connector (e.g., a USB connector) for transmitting/receiving power and/or data to and from an external electronic device. Although not shown, the electronic device **101** may include a plurality of connector holes **243**, and some of the plurality of connector holes **243** may function as connector holes for transmitting and receiving audio signals to and from an external electronic device. In the illustrated embodiment, the connector hole **243** is disposed on the 2-3.sup.th sidewall **221c**, but the disclosure is not limited thereto, and may be disposed on the 2-1.sup.th sidewall **221a** or the 2-2.sup.th second sidewall **221b**.

(61) According to one embodiment, the first housing **201** may include a first plate **211** and a slide cover (not shown). The first plate **211** and the slide cover (not shown) may be mounted (e.g., at least partially connected to) on the second housing **202**, and may move on the second housing **202** in a linear reciprocating motion in a direction (e.g., directions indicated by the arrow {circle around (1)} of FIG. 2B). According to one embodiment, the first plate **211** may support the flexible display **203**. For example, the first plate **211** may include a first face F1, and the first display area A1 of the flexible display **203** may be substantially positioned on the first face F1 to be maintained in the form of a flat panel. The slide cover may protect the flexible display **203** positioned on the first plate **211**. For example, at least a portion of the flexible display **203** may be positioned between the first plate **211** and the slide cover. According to one embodiment, the first plate **211** and the slide cover may be formed of a metal material and/or a non-metal (e.g., polymer) material. According to one embodiment, the first plate **211** may accommodate at least some (e.g., the battery **189** of FIG. 1, a motor (not shown), and a rack (not shown)) of the components of the electronic device **101**. According to various embodiments, at least one or a plurality of substrates may be accommodated in the second housing **202**. For example, the second housing **202** may include a circuit board (or a main circuit board) (not shown), and the main circuit board may include a processor, a memory, and/or an interface which are mounted thereon. The processor may include, for example, one or more of a central processing unit, an application processor, a graphics processing unit, an image signal processor, a sensor hub processor, or a communication processor. According to various embodiments, the circuit board may include a flexible printed circuit board type radio frequency cable (FRC). For example, the circuit board may be disposed on at least a portion of the second plate **221**, and may be electrically connected to an antenna module (e.g., the antenna module **197** of FIG. 1) and a communication module (e.g., the communication module **190** of FIG. 1).

(62) According to one embodiment, the memory may include, for example, a volatile memory or a non-volatile memory.

(63) According to one embodiment, the interface may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may, for example, electrically or physically connect the electronic device **101** to an external electronic device, and may include a USB connector, an SD card/MMC connector, or an audio connector.

(64) According to one embodiment, the electronic device **101** may further include a separate sub circuit board spaced apart from the circuit board in the second housing **202**. The sub circuit board may be electrically connected to a battery or electrical components disposed in the end area of the electronic device **101**, such as a speaker and/or a SIM socket, to transmit signals and power.

(65) According to one embodiment, the battery is a device configured to supply power to at least one element of the electronic device **101**, and may include, for example, a non-rechargeable primary battery or a rechargeable secondary battery, or a fuel cell. At least a portion of the battery may be disposed on, for example, substantially the same plane as the circuit board. The battery may be integrally disposed inside the electronic device **101**, or may be disposed to attachable to or detachable from the electronic device **101**.

(66) According to one embodiment, the electronic device **101** may include a motor (not shown) disposed on the second housing **202** (e.g., the second plate **221**) and configured to transmit a driving force to a rack (not shown) for guiding the sliding motion of the first housing **201** (e.g., the

first plate **211**) and a gear (not shown) for rotationally moving along the rack (not city).

(67) FIG. 3 illustrates a sensing structure for sensing expansion or reduction of a flexible display according to one embodiment.

(68) Referring to FIG. 3, the electronic device **101** according to one embodiment may include at least one hall sensor (e.g., a hall integrated chip (IC)) **320** in the second housing **202**, and a magnetic material (e.g., a magnet) **310** in the first housing **201**.

(69) The at least one hall sensor **320** according to one embodiment may include a plurality of hall sensors **322** to **328**, and the plurality of hall sensors **322** to **328** may be disposed at a predetermined interval in section {circle around (2)} in which the first housing **201** (or at least a portion of the flexible display **203**) moves in a reciprocating motion. Each of the plurality of hall sensors **322** to **328** may detect the strength (or change in the magnetic field) of the magnetic field and may provide information (or data or a value) (e.g., hall IC Raw data (X-axis, Y-axis, and Z-axis data)) of the detected magnetic field strength to a processor (e.g., the processor **120** of FIG. 1). For example, the magnetic field generated by the magnet **310** of the first housing **201** may be detected by the plurality of hall sensors **322** to **328** according to the movement of the first housing **201** (or at least a portion of the flexible display **203**) to a slide-out (an opened state) from a slide-in state (a closed state) with respect to the second housing **202**, or vice versa. The processor **120** according to one embodiment may identify (confirm or determinate or calculate) a distance (e.g., about several millimeters or centimeters) at which the first housing **201** (or at least a portion of the flexible display **203**) is slid out from the second housing **202**, based on the information of the magnetic field strength detected from each of the plurality of hall sensors **322** to **328**. For example, the processor **120** may identify, in units of about 1 mm, a distance at which the first housing **201** (or at least a portion of the flexible display **203**) is slid out from the second housing **202**. According to one embodiment, the plurality of hall sensors **322** to **328** has been described as including, for example, four sensors, but the number of hall sensors may be one or more and may not be limited to four sensors. According to one embodiment, an example of identifying a distance by at least one hall sensor and a magnet is given, but the distance at which the first housing **201** (or at least a part of the flexible display **203**) is slid out from the second housing **202** may also be identified by other known distance sensing methods. According to one embodiment, contrary to that shown in FIG. 3, the first housing **201** may include at least one hall sensor (e.g., a hall integrated chip) **320**, and the second housing **202** may include a magnetic material (e.g., a magnet) **310**.

(70) FIG. 4 is a block diagram of an electronic device according to an exemplary embodiment.

(71) Referring to FIG. 4, According to one embodiment, the electronic device **401** (e.g., the electronic device **101** of FIG. 1) may include at least one processor **420**, a memory **430**, a flexible display **460**, a slide driving unit **465**, a sensor module **476**, and/or a communication module **490**. According to one embodiment, the electronic device **401** is not limited thereto, and may be configured by further including various elements or by excluding some of the elements. According to one embodiment, the electronic device **401** may further include all or a part of the electronic device **101** illustrated in FIG. 1.

(72) According to one embodiment, the flexible display **460** (e.g., the display **160** of FIG. 1 or the flexible display **203** of FIGS. 2A to 2B) (hereinafter also referred to as a display) may be implemented in the form of a touch screen. When the flexible display **460** is implemented together with an input module in the form of a touch screen, the flexible display **460** may display various pieces of information generated according to a user's touch operation. According to one embodiment, the flexible display **460** may be configured to be rollable during a slide-in or slide-out operation. According to one embodiment, based on the slide-in or slide-out operation of the flexible display **460**, the flexible display **460** may enter a first state (e.g., an expanded state) in which a screen by the flexible display **460** is expanded or may enter a second state (e.g., a reduced state) in which the screen by the flexible display **460** is reduced. For example, the flexible display **460** may move in the second direction (a slide-out direction) with respect to the first housing **201** and/or the

second housing **202** to be in the expanded state (e.g., a partially expanded state or a fully expanded state) in which at least a portion thereof (e.g., **A2**) is exposed to the front face of the electronic device **101** from the inside of the first housing **201** and/or the second housing **202**, and may move in the first direction (e.g., a slide-in direction) opposite to the second direction with respect to the first housing **201** and/or the second housing **202** to be in the reduced state in which at least a portion thereof (e.g., **A2**) is not exposed to the front face of the electronic device **101**.

(73) According to one embodiment, the slide driving unit **465** may include a motor, a gear, and/or a rack, and the motor, the gear, and/or the rack may enable the flexible display **460** (e.g., the first housing **201**) (or at least a portion (e.g., **A2**) of the flexible display **203**) to be slid in or out.

According to one embodiment, the motor may be driven by receiving electric power (or current) to transmit the driving force to the gear connected to the motor. According to one embodiment, the motor may be implemented to provide a specified thrust (force) (e.g., a force of about 3 kgf) based on the repulsive force generated when the flexible display **460** slides in or out, and the motor may have a specified diameter (e.g., about 6.2 T or greater) to provide the specified force. According to one embodiment, the gear may be linked to the rack, based on the driving force from the motor such that the flexible display **460** slides in or out.

(74) According to one embodiment, the sensor module **476** (e.g., the sensor module **176** of FIG. **1**) may include at least one sensor. According to one embodiment, the at least one sensor may perform sensing to obtain a movement distance according to the movement of the first housing **201** (or at least a portion of the flexible display **460**) to the slide-out state (an opened state) from the slide-in state (a closed state) with respect to the second housing **202** or vice versa. For example, the at least one sensor may include at least one hall sensor (e.g., the hall IC **320** of FIG. **3**). According to one embodiment, the at least one hall sensor **320** may include a plurality of hall sensors **322** to **328**, and the plurality of hall sensors **322** to **328** may be disposed at a predetermined interval in section {circle around (2)} in which the first housing **201** (or at least a portion of the flexible display **203**) moves in a reciprocating motion. Each of the plurality of hall sensors **322** to **328** may detect the strength (or change in the magnetic field) of the magnetic field and may provide information (or data or a value) (e.g., hall IC Raw data (X-axis, Y-axis, and Z-axis data)) of the detected magnetic field strength to a processor **420**. For example, the magnetic field generated by the magnet **310** of the first housing **201** may be detected by the plurality of hall sensors **322** to **328** according to the movement of the first housing **201** (or at least a portion of the flexible display **203**) to the slide-out (an opened state) from the slide-in state (a closed state) with respect to the second housing **202**, or vice versa. According to one embodiment, the processor **120** may identify (confirm or determinate or calculate) a distance (e.g., about several millimeters or centimeters) at which the first housing **201** (or at least a portion of the flexible display **203**) is slid out from the second housing **202**, based on the information of the magnetic field strength detected from each of the plurality of hall sensors **322** to **328**. For example, the processor **420** may identify, in units of about 1 mm, a distance at which the first housing **201** (or at least a portion of the flexible display **460**) is slid out from the second housing **202**. According to one embodiment, as an example, the sensor module **476** includes a hall sensor, but the distance at which the first housing **201** (or at least a part of the flexible display **460**) is slid out from the second housing **202** may also be identified by other known distance sensing methods.

(75) According to one embodiment, The communication module **490** (e.g., the communication module **190** of FIG. **1**), may communicate with the external electronic device **104** through a first network **198** (e.g., a short-range communication network such as Bluetooth, wireless fidelity (WiFi) direct, or infrared data association (IrDA)) or a second network **199** (e.g., a telecommunications network such as a legacy cellular network, a 5G network, a next-generation communication network (e.g., new radio access technology), the Internet, or a computer network (e.g., a LAN or WAN)), based on the control of at least one processor **420**.

(76) According to one embodiment, the at least one processor **420** (e.g., the processor **120** of FIG.

1) may control overall operations of the elements included in the electronic device **401**.

(77) According to one embodiment, the processor **420** may identify whether the state of the flexible display **460** is an expanded state (e.g., a partially expanded state or a fully expanded state) or a reduced state. According to one embodiment, the processor **420** according to one embodiment may identify (confirm or determine or calculate) a distance (e.g., about several millimeters or centimeters) at which the first housing **201** (or at least a portion of the flexible display **460**) is slid out from the second housing **202**, by using information received from the sensor module **476** (e.g., at least one hall sensor **320**), and therefore, may identify whether the state of the flexible display **460** is an expanded state (e.g., a partially expanded state or a fully expanded state) or a reduced state. According to one embodiment, in the reduced state, the processor **420** may identify (or configure) the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** as a first input area (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment the processor **420** may display, on at least a portion of the first input area, information (e.g., a GUI guide or an image) indicating that it is an expansion (or slide-out) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is an expansion (or slide-out) request input area may be designated by a user. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters an expanded state, based on reception of a first input (e.g., a multi-touch input, a multi-swipe input, a swipe input after a single long touch, or inputs through other designated input methods) to the first input area (or the GUI guide in the first input area) in the reduced state. In the expanded state, according to one embodiment, the processor **420** may identify (or configure) the second edge area in the second direction (e.g., the expansion direction) of the flexible display **460** as a second input area (e.g., a slide-in (or reduction) input). According to one embodiment, the processor **420** may display, on at least a portion of the second input area, information (e.g., a GUI guide or an image) indicating that it is a reduction (or slide-in) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is a reduction (or slide-in) request input area may be designated by a user. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a reduced state, based on reception of a second input (e.g., a multi-touch input, a multi-touch swipe input, a swipe input after a single long touch, or inputs through other designated input methods) to the second input area (or the GUI guide in the second input area) in the expanded state.

(78) According to one embodiment, the processor **420** may identify whether the state of the flexible display **460** is a reduced state, a partially expanded state, or a fully expanded state.

(79) According to one embodiment, in the reduced state, the processor **420** may identify the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** as a first input area (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment, The processor **420** may display, on at least a portion of the first input area, information (e.g., a GUI guide or an image) indicating that it is an expansion (or slide-out) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is an expansion (or slide-out) request input area may be designated by a user. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or the expanded state, based on reception of the first input through the first input area (or the GUI guide in the first input area) in the reduced state. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or an expanded state, based on the type of the first input (e.g., a first multi-touch input (e.g., three-finger touch input) or a second multi-touch input (e.g., two-finger touch input)) received through the first input area in the reduced state. For

example, when the type of the first input received through the first input area in the reduced state is the first multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state, and when the type of the first input received through the first input area in the reduced state is the second multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a fully expanded state. According to one embodiment, when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or the expanded state, based on the type of running application (e.g., a content playback application (e.g., a media player application)) or the type of running file (e.g., video file or photo file). For example, in case that the type of the running application is a content playback application when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the ratio of the content playback screen, and in case that the type of the running application is not a content playback application when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a fully expanded state. For example, in case that the type of the running file is a video file or a photo file when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the aspect ratio of the video or photo, and in case that the type of the running file is not a video file or a photo file when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a fully expanded state. According to one embodiment, when the first input is received through the first input area in the reduced state, the processor **420** may display information (e.g., a pop-up menu) for selecting a partially expanded state or a fully expanded state, and may perform control such that the flexible display **460** enters a partially expanded state or the fully expanded state, based on the partially expanded state or the fully expanded state selected by a user.

(80) According to one embodiment, In the fully expanded state, the processor **420** may identify (or configure) the second edge area in the second direction (e.g., the expansion direction) of the flexible display **460** as a second input area (e.g., an input area for receiving reduction (or slide-in) request input). According to one embodiment, the processor **420** may display, on at least a portion of the second input area, information (e.g., a GUI guide or an image) indicating that it is a reduction (or slide-in) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is a reduction (or slide-in) request input area may be designated by a user. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or a reduced state, based on reception of the second input through the second input area (or the GUI guide in the second input area) in the fully expanded state. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or a reduced state, based on the type of the second input (e.g., a first multi-touch input (e.g., three-finger touch input) or a second multi-touch input (e.g., two-finger touch input)) received through the second input area in the fully expanded state. For example, when the type of the second input received through the second input area in the fully expanded state is the first multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state, and when the type of the second input received through the second input area in the fully expanded state is the second multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a reduced state. According to one embodiment, when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the flexible display **460** enters the partially expanded state or the reduced state, based on the type of running application (e.g., a content playback application (e.g., a media player application)) or the type of running file (e.g., video file or photo file). For example, in case that the type of the running

application is a content playback application when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the ratio of the content playback screen, and in case that the type of the running application is not a content playback application when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters a reduced state. For example, in case that the type of the running file is a video file or a photo file when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the aspect ratio of the video or photo, and in case that the type of the running file is not a video file or a photo file when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters the reduced state. According to one embodiment, when the second input is received through the second input area in the fully expanded state, the processor **420** may display information (e.g., a pop-up menu) for selecting a partially expanded state or a reduced state, and may perform control such that the flexible display **460** enters the partially expanded state or the reduced state, based on the partially expanded state or the reduced state selected by a user.

(81) According to one embodiment, in the partially expanded state, the processor **420** may identify (or configure) the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** as a first input area (e.g., an input area for receiving a reduction (or slide-in) request input), and may identify (or configure) a second edge area in the second direction (e.g., the expansion direction) as a second input area (e.g., an input area for receiving a reduction (or slide-in) request input). According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a fully expanded state, based on reception of the first input through the first input area (or the GUI guide in the first input area) in the partially expanded state. According to one embodiment, the processor **420** according to one embodiment may perform control such that the flexible display **460** enters a reduced state, based on reception of the second input through the second input area (or the GUI guide in the second input area) in the partially expanded state.

(82) According to one embodiment, the processor **420** may perform an operation by receiving the first input and/or the second input through the first input area and/or the second input area in a state in which the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** is designated as the first input area (e.g., an input area for receiving a reduction (or slide-in) request input) and the second edge area in the second direction (e.g., the expansion direction) is designated as the second input area (e.g., reduction (e.g., an input area for receiving a reduction (or slide-in) request input) regardless of whether the current state of the electronic device **401** is the reduced state, a partially expanded state, and/or a fully expanded state.

(83) According to one embodiment, the processor **420** may identify whether a display mode is a portrait view mode or a landscape view mode, identify the reduced state, the partially expanded state, and the fully expanded state in each of the landscape view mode and the portrait view mode, and perform an operation corresponding to each state.

(84) According to one embodiment, in the reduced state, the processor **420** may identify a third edge area the flexible display **460** disposed in a bottom direction (a direction to a bottom sidewall) from a center of the flexible display **460** as a first input area (e.g., an input area for receiving an expansion (or slide-out) request input), and in the expanded state, the third edge area may be identified as a second input area (e.g., an input area for receiving a reduction (or slide-in) request input).

(85) According to one embodiment, the memory **430** may store an application program and data. For example, the memory **430** may store an application (a function or a program) associated with a

reduction, partial expansion, or full expansion operation, based on identification of an input (e.g., an expansion request input or an expansion request input) and the identification of an input area (e.g., a first input area or a second input area) in the reduced state, the partially expanded state, and/or the fully expanded state of the flexible display **460**. According to one embodiment, the memory **430** according to one embodiment may store various data generated during execution of the program **140**, including a program (e.g., the program **140** of FIG. **1**) used for a functional operation. The memory **430** may largely include a program area **140** and a data area (not shown). The program area **140** may store related program information for driving the electronic device **401**, such as an operating system (OS) (e.g., the operating system **142** of FIG. **1**) for booting the electronic device **401**. The data area (not shown) may store transmitted and/or received data and generated data according to various embodiments. In addition, the memory **430** may be configured to include at least one storage medium of a flash memory, a hard disk, a multimedia card micro-type memory (e.g., a secure digital (SD) or extreme digital (XD) memory), RAM, and ROM.

(86) According to one embodiment, the electronic device **401** is not limited to the configuration shown in FIG. **4** and may further include various elements. According to one embodiment, the electronic device **401** may further include an image processing module (not shown). The image processing module may perform 2D or 3D image processing and/or rendering operation, based on the control of the processor **420**.

(87) In the above-described embodiment, the main elements of the electronic device have been described through the electronic device **401** of FIG. **4**. However, in various embodiments, not all of the elements shown in FIG. **4** are essential elements, and the electronic device **401** may be implemented by more elements than the illustrated elements, or the electronic device **401** may be implemented by fewer elements than the illustrated elements.

(88) According to various embodiments, the electronic device (e.g., the electronic device **101** of FIG. **1** or the electronic device **401** of FIG. **4**) may include a first housing (e.g., the first housing **201** of FIG. **2B**), a second housing (e.g., the second housing **202** of FIG. **2B**) configured to accommodate at least a portion of the first housing and connected to the first housing to enable sliding of the first housing, a flexible display (e.g., the display **160** of FIG. **1**, the flexible display **203** of FIG. **2B**, or the flexible display **460** of FIG. **4**) connected to the first housing, a slide driving unit (e.g., the slide driving unit **465** of FIG. **4**) configured to slidably move the first housing and the flexible display in an expansion direction or a reduction direction, and at least one processor (e.g., the processor **120** of FIG. **1** or the processor **420** of FIG. **4**), and the at least one processor may be configured to identify a reduced state in which the flexible display is slid in the reduction direction or an expanded state in which the flexible display is slid in the expansion direction, in the reduced state, identify a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and control the slide driving unit such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identify a second edge area of the flexible display disposed in the expansion direction from the center of the flexible display as a second input area and control the slide driving unit such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

(89) According to various embodiments, the electronic device may further include a magnet (e.g., the magnet **310** of FIG. **3**) disposed in the first housing, and at least one hall sensor (e.g., at least one hall sensor **320** of FIG. **3**) disposed in the second housing to sense a magnetic field generated by the magnet, and the at least one processor may be configured to identify, by using the at least one hall sensor, a distance at which the flexible display is slid in the reduction direction or the expansion direction.

(90) According to various embodiments, each of the first input and the second input may include one of a swipe input after single long touch, a multi-swipe input, or a multi-touch input.

(91) According to various embodiments, the at least one processor may be configured to display, on

at least a portion of the first edge area, first information indicating that the first edge area is an expansion request input area or display, on at least a portion of the second edge area, second information indicating that the second edge area is a reduction request input area.

(92) According to various embodiments, the at least one processor may be configured to designate a display position, size, color, or type of the first information or the second information, based on a user input.

(93) According to various embodiments, the at least one processor may be configured to identify whether the expanded state is a partially expanded state or a fully expanded state.

(94) According to various embodiments, the at least one processor may be configured to, when the first input is received through the first input area in the reduced state, identify whether the first input is a partial expansion request input or a full expansion request input, based on a designated condition, control the slide driving unit such that the flexible display enters a partially expanded state, based on identification of the partial expansion request input, and control the slide driving unit such that the flexible display enters a fully expanded state, based on identification of the full expansion request input.

(95) According to various embodiments, the designated condition may include one of a type of the first input, a type of an application to be executed upon the first input, a type of a file to be executed upon the first input, or a user selection input using a pop-up menu.

(96) According to various embodiments, the at least one processor may be configured to, when the second input is received through the second input area in the expanded state, identify whether the second input is a partial expansion request input or a reduction request input, based on a designated condition, control the slide driving unit such that the flexible display enters a partially expanded state, based on identification of the partial expansion request input, and control the slide driving unit such that the flexible display enters a reduced state, based on identification of the reduction request input.

(97) According to various embodiments, the at least one processor may be configured to, in the reduced state, identify a third edge area of the flexible display disposed in the bottom direction from a center of the electronic device as the first input area, and in the expanded state, identify the third edge area as the second input area.

(98) According to an embodiment of the disclosure, the electronic device (e.g., the electronic device **101** of FIG. **1** or the electronic device **401** of FIG. **4**) includes: (e.g., the first housing **201** of FIG. **2B**); a second housing (e.g., the second housing **202** of FIG. **2B**) configured to accommodate at least a portion of the first housing and including an open side through which the first housing is slidable; a flexible display (e.g., the display **160** of FIG. **1**, the flexible display **203** of FIG. **2B**, or the flexible display **460** of FIG. **4**) connected to the first housing; a slide driving unit (e.g., the slide driving unit **465** of FIG. **4**) configured to slidably move the first housing and the flexible display in an expansion direction or a reduction direction opposite the expansion direction; and at least one processor (e.g., the processor **120** of FIG. **1** or the processor **420** of FIG. **4**). and the at least one processor is configured to identify a reduced state in which the flexible display is slid in the reduction direction or an expanded state in which the flexible display is slid in the expansion direction, in the reduced state, identify a first edge area of the flexible display corresponding to a sidewall of the second housing as a first input area and control the slide driving unit such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identify a second edge area of the flexible display corresponding to an outermost sidewall of the first housing as a second input area and control the slide driving unit such that the flexible display enters the reduced state, based on reception of a second input through the second input area. The sidewall of the second housing is disposed opposite to the open side, and the outermost sidewall of the first housing is disposed opposite to the sidewall of the second housing.

(99) FIG. **5** is a flowchart showing an input area control operation for expanding and reducing a

display in an electronic device including a flexible display according to one embodiment.

(100) Referring to FIG. 5, the processor (e.g., the processor **120** of FIG. **1** or the processor **420** of FIG. **4**) of the electronic device (e.g., the electronic device **101** of FIG. **1** or the electronic device **401** of FIG. **4**) according to one embodiment may perform operations **510** to **570**.

(101) In operation **510**, according to one embodiment, the processor **420** may identify whether the state of the flexible display **460** is an expanded state (e.g., a partially expanded state or a fully expanded state) or a reduced state. According to one embodiment, the processor **420** may identify (confirm, determine, or calculate) a distance (e.g., about several millimeters or centimeters) at which the first housing **201** (or at least a portion of the flexible display **460**) is slid out from the second housing **202**, by using information received from the sensor module **476** (e.g., the at least one hall sensor **320**), and therefore, may identify whether the state of the flexible display **460** is an expanded state (e.g., a partially expanded state or a fully expanded state) or a reduced state.

(102) In operation **520**, according to one embodiment, in the reduced state, the processor **420** may configure the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** as a first input area (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment, the processor **420** may display, on at least a portion of the first input area, information (e.g., a GUI guide or an image) indicating that it is an expansion (or slide-out) request input area. For example, the display position, size, color, and/or type (text or image) which are indicating that it is an expansion (or slide-out) request input area may be designated by a user.

(103) In operation **530**, according to one embodiment, the processor **420** may identify reception of a first input (e.g., a multi-touch input, a multi-swipe input, a swipe input after a single long touch, or inputs through other designated input methods) to the first input area (or a GUI guide in the first input area) in the reduced state.

(104) In operation **540**, according to one embodiment, the processor **420** may perform expansion of a display. The processor **420** according to one embodiment may control the slide driving unit **465** such that the flexible display **460** enters an expanded state, based on identification (reception) of the first input in the first input area in the reduced state.

(105) In operation **550**, in the expanded state, according to one embodiment, the processor **420** may configure the second edge area in the second direction (e.g., the expansion direction) of the flexible display **460** as a second input area (e.g., an input area for receiving a slide-in (or reduction) input). According to one embodiment, the processor **420** according to one embodiment may display, on at least a portion of the second input area, information (e.g., a GUI guide or an image) indicating that it is a reduction (or slide-in) request input area. For example, the display position, size, color, and/or type (text or image), which are indicating that it is a reduction (or slide-in) request input area, may be designated by a user.

(106) In operation **560**, according to one embodiment, the processor **420** may identify reception of a second input (e.g., a multi-touch input, a multi-swipe input, a swipe input after a single long touch, or inputs through other designated input methods) to the second input area (or a GUI guide in the second input area) in the expanded state.

(107) In operation **570**, according to one embodiment, the processor **420** may perform reduction of a display. According to one embodiment, The processor **420** may control the slide driving unit **465** such that the flexible display **460** enters a reduced state, based on identification (reception) of the second input in the second input area.

(108) According to various embodiments, a method for controlling an input area for expansion or reduction of a display in the electronic device (e.g., the electronic device **101** of FIG. **1** or the electronic device **401** of FIG. **4**) may include identifying a reduced state in which the flexible display (e.g., the display **160** of FIG. **1** or the flexible display **460** of FIG. **4**) is slid in the reduction direction or an expanded state in which the flexible display is slid in the expansion direction, in the

reduced state, configuring a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and performing control such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, configuring a second edge area of the flexible display disposed in the expansion direction from the center of the flexible display as a second input area and performing control such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

(109) According to various embodiments, the method may include identifying a distance at which the flexible display is slid in the reduction direction or the expansion direction, by using a magnet (e.g., the magnet **310** of FIG. **3**) disposed in the first housing (e.g. the first housing **201** of FIGS. **2A** and **2B**) of the electronic device, and at least one hall sensor (e.g., at least one hall sensor **320** of FIG. **3**) disposed in the second housing (e.g. the second housing **202** of FIGS. **2A** and **2B**) of the electronic device to sense a magnetic field generated by the magnet.

(110) According to various embodiments, each of the first input and the second input may include one of a swipe input after single long touch, a multi-swipe input, or a multi-touch input.

(111) According to various embodiments, the method may further include displaying, on at least a portion of the first edge area, first information indicating that the first edge area is an expansion request input area or displaying, on at least a portion of the second edge area, second information indicating that the second edge area is a reduction request input area.

(112) According to various embodiments, the method may further include designating a display position, size, color, or type of the first information or the second information, based on a user input.

(113) According to various embodiments, the method may further include identifying whether the expanded state is a partially expanded state or a fully expanded state.

(114) According to various embodiments, the method may further include, when the first input is received through the first input area in the reduced state, identifying whether the first input is a partial expansion request input or a full expansion request input, based on a designated condition, and performing control such that the flexible display enters a partially expanded state, based on identification of the partial expansion request input or performing control such that the flexible display enters a fully expanded state, based on identification of the full expansion request input.

(115) According to various embodiments, the designated condition may include one of a type of the first input, a type of an application to be executed upon the first input, a type of a file to be executed upon the first input, or a user selection input using a pop-up menu.

(116) According to various embodiments, the method may further include, in the reduced state, identifying a third edge area of the flexible display disposed in the bottom direction from a center of the electronic device as the first input area, and in the expanded state, identifying the third edge area as the second input area.

(117) FIG. **6A** illustrates an example in which an electronic device transitions to an expanded state from a reduced state according to one embodiment. FIG. **6B** illustrates an example in which an electronic device transitions to a reduced state from an expanded state according to one embodiment.

(118) Referring to FIG. **6A**, according to one embodiment, the processor **420** may display a first display area **A1** in the reduced state <**601**>, and may configure a first edge area **610** of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** among the first display area **A1** as a first input area (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment, the processor **420** may control the slide driving unit **465** such that the flexible display **460** moves in the expansion direction (slide-out direction) with respect to the second housing **202**, based on a first input **612** (e.g., a multi-touch swipe input in the expansion direction) to the first input area **610**. When the flexible display **460** is moved the

maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a fully expanded state <**602**>, and when the flexible display **460** is moved less than the maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the fully expanded state <**602**>, at least a portion (e.g., **A2**) of the flexible display **460** may be exposed to the front face of the electronic device, and thus the display area may be expanded.

(119) Referring to FIG. **6B**, according to one embodiment, the processor **420** may perform display by using the first display area **A1** and the expanded display area **A2** in the expanded state (fully expanded state) <**602**>, and may configure the second edge area **620** in the second direction (e.g., the expansion direction) of the flexible display **460** as a second input area (e.g., an input area for receiving a reduction (or slide-in) request input). According to one embodiment, the processor **420** according to one embodiment may control the slide driving unit **465** such that the flexible display **460** moves in the reduction direction (slide-in direction) with respect to the second housing **202**, based on the second input **622** (e.g., a multi-touch swipe input in the reduction direction) to the second input area **620**. When the flexible display **460** is moved the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the display may be in a reduced state <**601**>, and when the flexible display **460** is moved less than the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the reduced state <**601**>, at least a portion (e.g., **A2**) of the flexible display **460** may not be exposed to the front face of the electronic device, and thus the display area may be reduced.

(120) FIG. **7A** illustrates GUI guide information displayed in a first input area in a reduced state of an electronic device according to one embodiment. FIG. **7B** illustrates GUI guide information displayed in a second input area in an expanded state of an electronic device according to one embodiment.

(121) Referring to FIG. **7A**, according to one embodiment, the processor **420** may display the first display area **A1** in the reduced state <**701**>, and in a state in which the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** among the first display area **A1** is configured as the first input area **610** (e.g., an input area for receiving an expansion (or slide-out) request input), may display, on at least a portion of the first input area, information (e.g., a GUI guide or an image) **710** indicating that it is an expansion (or slide-out) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is an expansion (or slide-out) request input area may be designated by a user. For example, the GUI guide **710** (e.g., an edge panel handle) may be displayed using at least a partial area of the first edge area **610** of the electronic device, and the position of at least the partial area on the first edge area **610** in which the edge panel handle is displayed may be changeable. According to one embodiment, The processor **420** may control the slide driving unit **465** such that the flexible display **460** moves in the expansion direction (slide-out direction) with respect to the second housing **202**, based on the first input **712** (e.g., a multi-touch swipe input in the expansion direction) to the GUI guide **710**. When the flexible display **460** is moved the maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a fully expanded state <**702**>, and when the flexible display **460** is moved less than the maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the fully expanded state <**702**>, at least a portion (e.g., **A2**) of the flexible display **460** may be exposed to the front face of the electronic device, and thus the display area may be expanded.

(122) Referring to FIG. **7B**, in a state in which the second edge area in the second direction (e.g.,

the expansion direction) of the flexible display **460** is identified as the second input area **620** (e.g., an input area for receiving a reduction (or slide-in) request input) in the expanded state (the fully expanded state) **<702>**, the processor **420** according to one embodiment may display, on at least a portion of the second input area **620**, information (e.g., a GUI guide or an image) **720** indicating that it is a reduction (or slide-in) request input area. For example, the GUI guide **720** (e.g., an edge panel handle) may be displayed using at least a partial area of the second edge area **620** of the electronic device, and the position of at least the partial area on the second edge area **620** in which the edge panel handle is displayed may be changeable. For example, the display position, size, color, and/or type (text or image) of information indicating that it is a reduction (or slide-in) request input area may be designated by a user. According to one embodiment, the processor **420** according to one embodiment may control the slide driving unit **465** such that the flexible display **460** moves in the reduction direction (slide-in direction) with respect to the second housing **202**, based on the second input **722** (e.g., a multi-touch swipe input in the reduction direction) to the GUI guide **720**. When the flexible display **460** is moved the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the flexible display **460** may be in a reduced state **<701>**, and when the flexible display **460** is moved less than the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the reduced state **<701>**, at least a portion (e.g., **A2**) of the flexible display **460** may not be exposed to the front face of the electronic device, and thus the display area may be reduced.

(123) FIG. **8A** illustrates an example in which an electronic device transitions to an expanded state from a reduced state when the electronic device is in a landscape view mode, according to one embodiment. FIG. **8B** illustrates an example in which an electronic device transitions to a reduced state from an expanded state when the electronic device is in a landscape view mode, according to one embodiment.

(124) Referring to FIG. **8A**, according to one embodiment, the processor **420** may display the horizontal screen (**A**) on the first display area **A1** while being in the landscape view mode and in the reduced state **<801>**, and may identify a first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** among the first display area **A1** as the first input area **610** (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment, the processor **420** may control the slide driving unit **465** such that the flexible display **460** moves in the expansion direction (slide-out direction) with respect to the second housing **202**, based on the first input **612** (e.g., a multi-touch swipe input in the expansion direction) to the first input area **610**. When the flexible display **460** is moved the maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a fully expanded state **<802>**, and when the flexible display **460** is moved less than the maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the fully expanded state **<802>**, at least a portion (e.g., **A2**) of the flexible display **460** may be exposed to the front face of the electronic device, and thus the landscape view display area may be expanded.

(125) Referring to FIG. **8B**, according to one embodiment, the processor **420** may perform display by using the first display area **A1** and the expanded display area **A2** while being in the landscape view mode and in the expanded state (the fully expanded state) **<802>**, and may identify the second edge area in the second direction (e.g., the expansion direction) of the flexible display **460** as the second input area **620** (e.g., an input area for receiving a reduction (or slide-in) request input). According to one embodiment, the processor **420** according to one embodiment may control the slide driving unit **465** such that the flexible display **460** moves in the reduction direction (slide-in direction) with respect to the second housing **202**, based on the second input **622** (e.g., a multi-

touch swipe input in the reduction direction) to the second input area **620**. When the flexible display **460** is moved the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the flexible display **460** may be in a reduced state **<801>**, and when the flexible display **460** is moved less than the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the reduced state **<801>**, at least a portion (e.g., **A2**) of the flexible display **460** may not be exposed to the front face of the electronic device, and thus the landscape view display area may be reduced.

(126) FIG. **9** illustrates an example in which a first input or a second input designated to be identified in a first input area or a second input area of an electronic device is a swipe input after a long touch, according to one embodiment.

(127) Referring to FIG. **9**, when a swipe input after a long touch is received as the first input **912** in the first input area **610** in the reduced state **<901>**, the processor **420** according to one embodiment may control the slide driving unit **465** such that the flexible display **460** moves in the expansion direction (slide-out direction) with respect to the second housing **202**. When a swipe input after a long touch is received as the second input **922** in the second input area **620** in the expanded state **<902>**, according to one embodiment the processor **420** may control the slide driving unit **465** such that the flexible display **460** moves in the reduction direction (slide-in direction) with respect to the second housing **202**.

(128) FIG. **10** is a flowchart showing an input area control operation for full expansion, partial expansion, and reduction of a display in an electronic device according to one embodiment.

(129) Referring to FIG. **10**, according to one embodiment, the processor (e.g., the processor **120** of FIG. **1** or the processor **420** of FIG. **4**) of the electronic device (e.g., the electronic device **101** of FIG. **1** or the electronic device **401** of FIG. **4**) may perform at least one of operations **1005** to **1090**.

(130) In operation **1005**, according to one embodiment, the processor **420** may identify whether the state of the flexible display **460** is a reduced state, a partially expanded state, or a fully expanded state. According to one embodiment, the processor **420** may identify (confirm or determine or calculate) a distance (e.g., about several millimeters or centimeters) at which the first housing **201** (or at least a portion of the flexible display **460**) is slid out from the second housing **202**, by using information received from the sensor module **476** (e.g., at least one hall sensor **320**), and therefore, may identify whether the state of the flexible display **460** is a reduced state, a partially expanded state, or a fully expanded state.

(131) In operation **1010**, according to one embodiment, the processor **420** may identify, in the reduced state, the first edge area in a first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) of the flexible display **460** as a first input area (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment, the processor **420** may display, on at least a portion of the first input area, information (e.g., a GUI guide or an image), indicating that it is an expansion (or slide-out) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is an expansion (or slide-out) request input area may be designated by a user.

(132) In operation **1020**, according to one embodiment, the processor **420** may identify (or receive) a first input through the first input area (or a GUI guide in the first input area) in the reduced state.

(133) In operation **1030**, according to one embodiment, the processor **420** may perform control such that the flexible display **460** enters an expanded state (a partially expanded state or a fully expanded state), based on identification of the first input through the first input area in the reduced state. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or an expanded state, based on the type of the first input (e.g., a first multi-touch input (e.g., three-finger touch input) or a second multi-touch input (e.g., two-finger touch input)) received through the first input area in the reduced state. For example, when the type of the first input received through the first input area in the reduced state is

the first multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state, and when the type of the first input received through the first input area in the reduced state is the second multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a fully expanded state. According to one embodiment, when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or the expanded state, based on the type of running application (e.g., a content playback application (e.g., a media player application)) or the type of running file (e.g., video file or photo file). For example, in case that the type of the running application is a content playback application when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the ratio of the content playback screen, and in case that the type of the running application is not a content playback application when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a fully expanded state. For example, in case that the type of the running file is a video file or a photo file when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the aspect ratio of the video or photo, and in case that the type of the running file is not a video file or a photo file when the first input is received through the first input area in the reduced state, the processor **420** may perform control such that the display enters a fully expanded state. According to one embodiment, when the first input is received through the first input area in the reduced state, the processor **420** may display information (e.g., a pop-up menu) for selecting a partially expanded state or a fully expanded state, and may perform control such that the flexible display **460** enters a partially expanded state or the fully expanded state, based on the partially expanded state or the fully expanded state selected by a user.

(134) In operation **1040**, according to one embodiment, in the fully expanded state, the processor **420** may identify the second edge area in the second direction (e.g., the expansion direction) of the flexible display **460** as a second input area (e.g., an input area for receiving reduction (or slide-in) request input). According to one embodiment, the processor **420** may display, on at least a portion of the second input area, information (e.g., a GUI guide or an image) indicating that it is a reduction (or slide-in) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is a reduction (or slide-in) request input area may be designated by a user.

(135) In operation **1050**, according to one embodiment, the processor **420** may identify (or receive) the second input through the second input area (or a GUI guide in the second input area) in the fully expanded state.

(136) In operation **1060**, according to one embodiment, the processor **420** according to one embodiment may perform control such that the flexible display **460** is reduced (a partially expanded state or a reduced state), based on reception of the second input through the second input area in the fully expanded state. According to one embodiment, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state or a reduced state, based on the type of the second input (e.g., a first multi-touch input (e.g., three-finger touch input) or a second multi-touch input (e.g., two-finger touch input)) received through the second input area in the fully expanded state. For example, when the type of the second input received through the second input area in the fully expanded state is the first multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a partially expanded state, and when the type of the second input received through the second input area in the fully expanded state is the second multi-touch input, the processor **420** may perform control such that the flexible display **460** enters a reduced state. According to one embodiment, when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that

the flexible display **460** enters the partially expanded state or the reduced state, based on the type of running application (e.g., a content playback application (e.g., a media player application)) or the type of running file (e.g., video file or photo file). For example, in case that the type of the running application is a content playback application when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the ratio of the content playback screen, and in case that the type of the running application is not a content playback application when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters a reduced state. For example, in case that the type of the running file is a video file or a photo file when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters a partially expanded state corresponding to the aspect ratio of the video or photo, and in case that the type of the running file is not a video file or a photo file when the second input is received through the second input area in the fully expanded state, the processor **420** may perform control such that the display enters the reduced state. According to one embodiment, when the second input is received through the second input area in the fully expanded state, the processor **420** may display information (e.g., a pop-up menu) for selecting a partially expanded state or a reduced state, and may perform control such that the flexible display **460** enters the partially expanded state or the reduced state, based on the partially expanded state or the reduced state selected by a user.

(137) In operation **1070**, according to one embodiment, in the partially expanded state, the processor **420** may identify the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** as a first input area (e.g., an input area for receiving a reduction (or slide-in) request input), and may identify a second edge area in the second direction (e.g., the expansion direction) as a second input area (e.g., an input area for receiving a reduction (or slide-in) request input).

(138) In operation **1080**, according to one embodiment, in the partially expanded state, the processor **420** may identify (or receive) a first input through the first input area or identify (or receive) a second input through the second input area.

(139) In operation **1090**, according to one embodiment, in the partially expanded state, the processor **420** may perform control such that the flexible display **460** enters a fully expanded state, based on identification of the first input through the first input area, and may perform control such that the flexible display **460** enters a reduced state, based on identification of the second input through the second input area.

(140) FIG. **11** illustrates a reduced state, a partially expanded state, and a fully expanded state of a display in an electronic device according to one embodiment.

(141) Referring to FIG. **11**, according to one embodiment, the processor **420** may perform control such that the flexible display **460** enters one of a partially expanded state <**1102**> and a fully expanded state <**1103**>, based on a designated condition (e.g., a type of the first input, a type of an application to be executed upon the first input, a type of a file to be executed upon the first input, or a user selection input using a pop-up menu) when the first input **1112** is identified through the first input area **1110** in the reduced state <**1101**>. For example, at least a partial area (A2') of the display exposed based on the partial expansion in the partially expanded state <**1102**> may be a smaller than at least a partial area A2 of the display exposed based on the full expansion in the fully expanded state <**1103**>.

(142) According to one embodiment, the processor **420** according to one embodiment may perform control such that the flexible display **460** enters a fully expanded state <**1103**> when the first input **1112** is identified through the first input area **1110** in the partially expanded state <**1101**>, and may perform control such that the flexible display **460** enters a reduced state <**1101**> when the second input **1122** is identified through the second input area **1120**.

(143) According to one embodiment, the processor **420** according to one embodiment may perform control such that the flexible display **460** enters one of a partially expanded state <**1102**> and a reduced state <**1101**>, based on a designated condition (e.g., a type of the second input, a type of an application to be executed upon the second input, a type of a file to be executed upon the second input, or a user selection input using a pop-up menu) when the second input **1122** is identified through the second input area **1120** in the full expanded state <**1103**>.

(144) FIG. **12** illustrates an example in which a first edge area in a first direction of a display is designated as a first input area, and a second edge area in a second direction is designated as a second input area of a display according to one embodiment.

(145) Referring to FIG. **12**, according to one embodiment, the processor **420** according to one embodiment may perform operation by receiving the first input **1212** and/or the second input **1222** through the first input area **1210** and/or the second input area **1220** in a state in which the first edge area of the flexible display **460** disposed in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) from a center of the flexible display **460** is designated as the first input area (e.g., an input area for receiving a reduction (or slide-in) request input), and the second edge area in the second direction (e.g., the expansion direction) is designated as the second input area (e.g., an input area for receiving a reduction (or slide-in) request input) regardless of whether the current state of the electronic device **401** is the reduced state, a partially expanded state, and/or a fully expanded state.

(146) FIG. **13** illustrates an example in which an electronic device enters a partially expanded state or a fully expanded state during display of a content while the electronic device is in a reduced state and in a landscape view mode, according to one embodiment.

(147) Referring to FIG. **13**, according to one embodiment, the processor **420** may identify a first input **1312** through a first input area **1310** during display of a content (e.g., a video) **1362** on the first display area **A1** while being in a landscape view mode and in a reduced state <**1301**>. For example, the first input area **1310** may be an area configured (or designated) on a portion or entire portion of the first edge area in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) in the landscape view mode and in a reduced state <**1301**>. For example, the first input **1312** may include a two-finger swipe down input to the first input area **1310**. For example, the first input **1312** may be a gesture input other than a two-finger swipe down input.

(148) According to one embodiment, based on an aspect ratio of a content **1362** when the first input **1312** is identified while the content **1362** is displayed on the first display area **A1** in the landscape view mode and in reduced state <**1301**>, the processor **420** may perform control such that the state of the display enters a partially expanded state <**1302**> corresponding to (or suitable for) the aspect ratio, and may display the content **1362** in the display area (**A1**+**A2'**) corresponding to the partially expanded state. According to one embodiment, the processor **420** may identify the second input **1324** through the second input area **1320** in the landscape view mode and in the partially expanded state <**1302**>. For example, the second input area **1320** may be an area configured (or designated) on a portion or entire portion of the second edge area in the first direction (e.g., the fixing direction, the opposite direction to the expansion direction, or the reduction direction) in the landscape view mode and in the partially expanded state <**1302**>. For example, as the reduced state <**1301**> becomes the partially expanded state <**1302**>, the first edge area and the second edge area may be different. For example, the second input **1324** may include a two-finger swipe down input. For example, the second input **1324** may be a gesture input other than a two-finger swipe down input. According to one embodiment, when the second input is received through the second input area **1320** in the partially expanded state <**1302**> and in the landscape view mode, the processor **420** receives may perform control such that the display enters a fully expanded state <**1303**>, and may display the content **1362** in the display area **A1**+**A2** corresponding to the fully expanded state.

(149) Although it has been described with reference to the example of FIG. **13** that after expansion

from the reduced state <1301> in the landscape view mode to the partially expanded state <1302>, based on the first input 1312, the processor 420 causes the partially expanded state <1302> to be the fully expanded state <1303>, based on the second input 1324, the operation of the partially expanded state <1302> may be omitted depending on the configuration. For example, the processor 420 may cause the reduced state <1301> in the landscape view mode to be the fully expanded state <1303> without the partially expanded state <1302>, based on a third input (not shown). For example, the third input may include a three-finger swipe down input. For example, the third input may be a different type of gesture input different from the first input, other than the three-finger swipe down input.

(150) According to one embodiment, the processor 420 may select whether to operate to cause the reduced state <1301> in the landscape view mode to be the fully expanded state <1303> after the partially expanded state <1302> or to be the fully expanded state <1303> without the partially expanded state <1302>, based on whether an application is designated (or designated application type) or whether a file type is designated or based on user selection. For example, the designated application may include a media playback application or an over the top (OTT) application. For example, the designated file type may include a video file type file or a photo file type. For example, the user selection may include a partial expansion or full expansion selection using a pop-up menu, and when user selection is used, the processor 420 may display a pop-up menu, based on reception of an expansion command execution input in a reduced state, and the expansion may be performed according to the partial expansion or the full expansion selected by a user among the partial expansion or the full expansion included in the pop-up menu.

(151) FIG. 14 illustrates an example of utilizing an expanded area when an electronic device transitions to a partially expanded state or a fully expanded state from a reduced state according to one embodiment.

(152) Referring to FIG. 14, according to one embodiment, the processor 420 may cause such that when the reduced state <1401> is changed to the partially expanded state <1402> during display of a first content (e.g., a first preview image) 1464 in the first display area A1, the display area is enlarged due to the expanded area (A2'), and accordingly, a second content (e.g., a menu associated with the first preview image) 1465 may be further displayed. According to one embodiment, the processor 420 according to one embodiment may cause such that when the partially expanded state <1402> is changed to the fully expanded state <1403>, the display has an area wider than that in the partially expanded state <1402> due to the expanded area A2. According to one embodiment, the processor 420 according to one embodiment may cause such that display displays, in the fully expanded state <1403>, the second content (e.g., a second preview image) 1466 having a screen size larger than that of the first content 1464.

(153) FIG. 15 is a flowchart showing a control operation when a third edge area at the lower end of a display is used as an input area for expanding or reducing the display in an electronic device according to one embodiment.

(154) Referring to FIG. 15, according to one embodiment, the processor (e.g., the processor 120 of FIG. 1 or the processor 420 of FIG. 4) of the electronic device (e.g., the electronic device 101 of FIG. 1 or the electronic device 401 of FIG. 4) may perform at least one operation among the operations 1510 to 1570.

(155) In operation 1510, according to one embodiment, the processor 420 may identify whether the state of the flexible display 460 is an expanded state (e.g., a partially expanded state or a fully expanded state) or a reduced state. According to one embodiment the processor 420 according to one embodiment may identify (confirm or determine or calculate) a distance (e.g., about several millimeters or centimeters) at which the first housing 201 (or at least a portion of the flexible display 460) is slid out from the second housing 202, by using information received from the sensor module 476 (e.g., at least one hall sensor 320), and therefore, may identify whether the state of the flexible display 460 is an expanded state (e.g., a partially expanded state or a fully expanded

state) or a reduced state.

(156) In operation **1520**, according to one embodiment, in the reduced state, the processor **420** may identify the third edge area at the lower end of display **460** as an expansion input area (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment, the processor **420** may display, on at least a portion of the expansion input area, information (e.g., a GUI guide or an image) indicating that it is an expansion (or slide-out) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is an expansion (or slide-out) request input area may be designated by a user.

(157) In operation **1530**, according to one embodiment, the processor **420** may identify (or receive) a first input ((e.g., a multi-touch input, a multi-swipe input, a swipe input after a single long touch, or inputs through other designated input methods) through the third edge area of the lower end of the display in the reduced state.

(158) In operation **1540**, according to one embodiment, the processor **420** may perform expansion of a display. According to one embodiment, the processor **420** may control the slide driving unit **465** such that the flexible display **460** enters an expanded state, based on identification (reception) of the first input through the third edge area at the lower end of the display in the reduced state.

(159) In operation **1550**, according to one embodiment in the expanded state, the processor **420** may identify the third edge area at the lower end of display **460** as a reduction input area (e.g., an input area for receiving a reduction (or slide-in) request input). According to one embodiment, the processor **420** may display, on at least a portion of the reduction input area, information (e.g., a GUI guide or an image) indicating that it is a reduction (or slide-in) request input area. For example, the display position, size, color, and/or type (text or image) of information indicating that it is a reduction (or slide-in) request input area may be designated by a user.

(160) In operation **1560**, according to one embodiment, the processor **420** may identify (or receive) a second input ((e.g., a multi-touch input, a multi-swipe input, a swipe input after a single long touch, or inputs through other designated input methods) through the third edge area in the expanded state.

(161) In operation **1570**, according to one embodiment, the processor **420** may perform reduction of a display. According to one embodiment, the processor **420** may control the slide driving unit **465** such that the flexible display **460** enters a reduced state, based on identification (reception) of the second input through the third edge area.

(162) FIG. **16** illustrates an example of receiving a first input or a second input through a third edge area at the lower end of a display according to one embodiment.

(163) Referring to FIG. **16**, according to one embodiment, the processor **420** may display a first display area **A1** in the reduced state <**1601**>, and may identify a third edge area **1610** at the lower end of the flexible display **460** among the first display area **A1** as the first input area (e.g., an input area for receiving an expansion (or slide-out) request input). According to one embodiment, the processor **420** may control the slide driving unit **465** such that the flexible display **460** moves in the expansion direction (slide-out direction) with respect to the second housing **202**, based on a first input **1612** (e.g., a multi-touch swipe input in the top direction) to the first input area **1610**. When the flexible display **460** is moved the maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a fully expanded state <**1602**>, and when the flexible display **460** is moved less than the maximum movable distance in the expansion direction (slide-out direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the fully expanded state <**1602**>, at least a portion (e.g., **A2**) of the flexible display **460** may be exposed to the front face of the electronic device, and thus the display area may be expanded.

(164) Referring to FIG. **16**, according to one embodiment, the processor **420** may perform display by using the first display area **A1** and the expanded display area **A2** in the expanded state (fully expanded state) <**1602**>, and may identify the third edge area at the lower end of the flexible

display **460** as a second input area **1620** (e.g., an input area for receiving a reduction (or slide-in) request input). According to one embodiment, the processor **420** according to one embodiment may control the slide driving unit **465** such that the flexible display **460** moves in the reduction direction (slide-in direction) with respect to the second housing **202**, based on the second input **1622** (e.g., a multi-touch swipe input in the top direction) to the second input area **1620**. When the flexible display **460** is moved the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the display may be in a reduced state <**1601**>, and when the flexible display **460** is moved less than the maximum movable distance in the reduction direction (slide-in direction) with respect to the second housing **202**, the flexible display **460** may be in a partially expanded state. In the reduced state <**1601**>, at least a portion (e.g., A2) of the flexible display **460** may not be exposed to the front face of the electronic device, and thus the display area may be reduced.

(165) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(166) It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C”, may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(167) As used in connection with the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic”, “logic block”, “part”, or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(168) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage

medium and where the data is temporarily stored in the storage medium.

(169) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(170) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. According to various embodiments, one or more of the above-described components or operations may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(171) According to various embodiments, a non-volatile storage medium storing instructions which, when executed, cause an electronic device comprising at least one processor to perform at least one operation, and the at least one operation may include identifying a reduced state in which the flexible display is slid in the reduction direction or an expanded state in which the flexible display is slid in the expansion direction, in the reduced state, identifying a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and performing control such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identifying a second edge area of the flexible display disposed in the expansion direction from the center of the flexible display as a second input area and performing control such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

(172) The embodiments of the disclosure described and shown in the specification and the drawings are merely specific examples that have been presented to easily explain the technical contents of the disclosure and help understanding of the disclosure, and are not intended to limit the scope of the disclosure. Therefore, the scope of the disclosure should be construed to include, in addition to the embodiments disclosed herein, all changes and modifications derived on the basis of the technical idea of the disclosure.

Claims

1. An electronic device comprising: a first housing; a second housing configured to accommodate at least a portion of the first housing and connected to the first housing to enable sliding of the first housing; a flexible display connected to the first housing; a slide driving unit configured to slidably move the first housing and the flexible display in an expansion direction or a reduction direction opposite to the expansion direction; and at least one processor, wherein the at least one processor is configured to, identify a reduced state in which the flexible display is slid in the reduction direction or an expanded state in which the flexible display is slid in the expansion direction, in the reduced state, identify a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and control the slide

driving unit such that the flexible display enters the expanded state, based on reception of a first input through the first input area, and in the expanded state, identify a second edge area of the flexible display disposed in the expansion direction from the center of the flexible display as a second input area and control the slide driving unit such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

2. The electronic device of claim 1, further comprising: a magnet disposed in the first housing; and at least one hall sensor disposed in the second housing to sense a magnetic field generated by the magnet, wherein the at least one processor is configured to identify, by using the at least one hall sensor, a distance at which the flexible display is slid in the reduction direction or the expansion direction.

3. The electronic device of claim 1, wherein each of the first input and the second input comprises one of a swipe input after single long touch, a multi-swipe input, or a multi-touch input.

4. The electronic device of claim 1, wherein the at least one processor is configured to display, on at least a portion of the first edge area, first information indicating that the first edge area is an expansion request input area or display, on at least a portion of the second edge area, second information indicating that the second edge area is a reduction request input area.

5. The electronic device of claim 4, wherein the at least one processor is configured to designate a display position, size, color, or type of the first information or the second information, based on a user input.

6. The electronic device of claim 1, wherein the at least one processor is configured to identify whether the expanded state is a partially expanded state or a fully expanded state.

7. The electronic device of claim 1, wherein the at least one processor is configured to, when the first input is received through the first input area in the reduced state, identify whether the first input is a partial expansion request input or a full expansion request input, based on a designated condition, control the slide driving unit such that the flexible display enters a partially expanded state, based on identification of the partial expansion request input, and control the slide driving unit such that the flexible display enters a fully expanded state, based on identification of the full expansion request input.

8. The electronic device of claim 7, wherein the designated condition comprises one of a type of the first input, a type of an application to be executed upon the first input, a type of a file to be executed upon the first input, or a user selection input using a pop-up menu.

9. The electronic device of claim 1, wherein the at least one processor is configured to, when the second input is received through the second input area in the expanded state, identify whether the second input is a partial expansion request input or a reduction request input, based on a designated condition, control the slide driving unit such that the flexible display enters a partially expanded state, based on identification of the partial expansion request input, and control the slide driving unit such that the flexible display enters the reduced state, based on identification of the reduction request input.

10. The electronic device of claim 1, wherein the at least one processor is configured to, in the reduced state, identify a third edge area of the flexible display disposed in a bottom direction from a center of the electronic device as the first input area, and in the expanded state, identify the third edge area as the second input area.

11. A method for controlling an input area for expansion or reduction of a display in an electronic device, the method comprising: identifying a reduced state in which a flexible display is slid in a reduction direction or an expanded state in which the flexible display is slid in an expansion direction, wherein the expansion direction is opposite to the reduction direction; in the reduced state, identifying a first edge area of the flexible display disposed in the reduction direction from a center of the flexible display as a first input area and performing control such that the flexible display enters the expanded state, based on reception of a first input through the first input area; and in the expanded state, identifying a second edge area of the flexible display disposed in the

expansion direction from the center of the flexible display as a second input area and performing control such that the flexible display enters the reduced state, based on reception of a second input through the second input area.

12. The method of claim 11, further comprising: identifying a distance at which the flexible display is slid in the reduction direction or the expansion direction, by using a magnet disposed in a first housing of the electronic device, and at least one hall sensor disposed in a second housing of the electronic device to sense a magnetic field generated by the magnet.

13. The method of claim 11, wherein each of the first input and the second input comprises one of a swipe input after single long touch, a multi-swipe input, or a multi-touch input.

14. The method of claim 11, further comprising: displaying, on at least a portion of the first edge area, first information indicating that the first edge area is an expansion request input area or displaying, on at least a portion of the second edge area, second information indicating that the second edge area is a reduction request input area.

15. The method of claim 14, further comprising: designating a display position, size, color, or type of the first information or the second information, based on a user input.

16. The method of claim 11, further comprising: identifying whether the expanded state is a partially expanded state or a fully expanded state.

17. The method of claim 11, further comprising: when the first input is received through the first input area in the reduced state, identifying whether the first input is a partial expansion request input or a full expansion request input, based on a designated condition; and performing control such that the flexible display enters a partially expanded state, based on identification of the partial expansion request input or performing control such that the flexible display enters a fully expanded state, based on identification of the full expansion request input.

18. The method of claim 17, wherein the designated condition comprises one of a type of the first input, a type of an application to be executed upon the first input, a type of a file to be executed upon the first input, or a user selection input using a pop-up menu.

19. The method of claim 11, further comprising: in the reduced state, identifying a third edge area of the flexible display disposed in a bottom direction from a center of the electronic device as the first input area; and in the expanded state, identifying the third edge area as the second input area.

20. A non-volatile storage medium that stores instructions which, when executed by an electronic device, cause the electronic device to perform at least one operation, the at least one operation comprising: identifying a reduced state in which a flexible display is slid in a reduction direction or an expanded state in which the flexible display is slid in an expansion direction, wherein the expansion direction is opposite to the reduction direction; in the reduced state, identifying a first edge area of the flexible display in the reduction direction from a center of the flexible display as a first input area and performing control such that the flexible display enters the expanded state, based on reception of a first input through the first input area; and in the expanded state, identifying a second edge area of the flexible display in the expansion direction from a center of the flexible display as a second input area and performing control such that the flexible display enters the reduced state, based on reception of a second input through the second input area.
